

Jordan Vega

Robotics & Embedded Systems Engineer

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Professional Summary

Robotics and embedded systems engineer with hands-on experience deploying, repairing, validating, and integrating autonomous robotic and industrial hardware systems in real-world environments. Skilled in ROS2, embedded Linux, hardware validation, sensor fusion, industrial communications, and full-stack system integration across software, electronics, networking, and control layers. Proven ability to diagnose failures and restore systems under live operational pressure. Targeting aerospace, defense, robotics, and embedded autonomy roles.

Technical Skills

Autonomy & Robotics: ROS2 Humble, PX4 Autopilot (SITL), MAVLink, MAVSDK, SLAM, Kalman filtering, obstacle avoidance

Perception: OpenCV, ArUco tracking, Intel RealSense D455, YOLOv8 (Jetson TensorRT), depth sensing

Embedded & Electronics: Raspberry Pi 4B, NVIDIA Jetson Orin Nano, STM32, PIC32, FPGA tooling, UART, SPI, I2C, PWM, GPIO

Electrical Test & Validation: Multimeters, optical power meters, programmable electronic loads, DC-DC converter testing, relay logic validation, continuity testing

Systems & Networking: Linux (Ubuntu 22.04), Docker, Tailscale VPN, WebSockets, TCP/UDP, distributed systems

Programming: Python, C++, Go, TypeScript, JavaScript, Bash

Tools: Git, VS Code, FastAPI, Next.js, React, CycloneDDS, MPLAB ICD3, Quartus

Professional Experience

Electronics Test & Validation Technician Apr 2026 â Present

Meridian Test Labs, Inc. â Fairfield, CA

â Performed validation, calibration, and troubleshooting of fiber optic communication modules, verifying electrical & optical â electrical signal integrity across telecom and industrial communication systems

â Measured voltage, current, resistance, and continuity using multimeters, programmable electronic loads, and bench power supplies

â Validated optical transmission systems at 850nm and 1310nm wavelengths using calibrated optical power meters & dBm measurements

â Executed DC-DC power converter testing and calibration including voltage regulation, load testing (â95â100W), tolerance verification ($\pm 0.5\%$), and thermal drift observation

â Calibrated relay threshold detection and alarm logic circuits (VR1/VR2), validating LED indicators, switching behavior, and fault conditions

â Performed firmware flashing, embedded hardware bring-up, and web interface deployment for industrial Ethernet and I/O systems using MPLAB ICD3, PIC32 microcontrollers, MPFS2, USB Blaster, and Quartus

â Conducted board-level troubleshooting and failure isolation using continuity testing, embedded network diagnostics, and known-good reference hardware ("golden boards")

â Assisted with FPGA programming, board-level validation, and integration testing for industrial communication and supervisory control systems

Robotics Deployment Technician Jan 2025 â Dec 2025

Vanguard Robotics â Hollywood, CA

â Maintained 24/7 deployment of an autonomous robotic system in a high-volume production environment as the on-site technical owner

â Assembled and integrated new hardware components to improve system autonomy and operational reliability

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â Performed repairs across mechanical, electrical, and control systems during live robotic operation

â Diagnosed and resolved failures across hardware, perception, and control systems under real-time constraints

â Replaced and calibrated robotic grippers on industrial robotic arms and validated performance under production

â Calibrated vision systems using ArUco markers and supported live iteration with engineering teams

â Acted as liaison between operators, staff, and engineers during system failures and recovery